

Priority-Dependent Display of Notifications in the Peripheral Field of View of Smart Glasses

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ABSTRACT

We propose a concept for displaying notifications in the peripheral field of view of smart glasses aiming to achieve a balance between perception and distraction depending on the priority of the notification. We designed three different visualizations for notifications of low, medium, and high priority. To evaluate this concept, we conducted a study with 24 participants who reacted to the notifications while performing a primary task. Reaction times for the low-priority notification were significantly higher. The medium- and high-priority notifications did not show a clear difference.

Keywords: Augmented reality glasses, notification, interruption, distraction, peripheral vision.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

Smart glasses are becoming increasingly commercialized with lower weights and compact form factors. Thus, they are more comfortable to wear and more suitable for daily use. In this context, researchers have suggested that smart glasses could replace or at least complement smartphones someday [1]. Common features of mobile devices, such as notifications, should then also be available for smart glasses. Notifications supply users with information outside of their current attentional focus via visual, auditory, or haptic cues [2]. Users receive around 100 notifications throughout a day interrupting and distracting them during primary tasks [3]. Some of the notifications contain important or urgent information while others are less urgent and can even be ignored [4]. Interruptions due to less important notifications may not be desirable. Hence, the objective of this study is to achieve a balance between the disruptive character of notifications and their informational benefit.

Displaying notifications on smart glasses involves certain advantages in this context. Users can access information quickly and do not need to take their smartphones out of their pockets. They can stay engaged in their primary task while perceiving the notifications and this requires less abrupt attentional shifts [5]. Additionally, the hands remain free to perform other tasks, which may also reduce the disruptiveness of notifications. Even the central field of view of smart glasses can be kept free if notifications are displayed in the peripheral areas. This is particularly relevant for mobile dual-task contexts such as while



Figure 1: Notification icons for the categories low priority (left), medium priority (middle), and high priority (right).

riding a bike [6]. However, research regarding notifications on smart glasses is still limited and further studies are required.

The present study, therefore, aims to provide further insights regarding notifications on smart glasses. More precisely, we developed a concept for notifications presented in the peripheral field of view of smart glasses. The concept includes three visualizations representing notifications of low, medium, and high priority. We thereby aimed at achieving the previously mentioned balance between disruptiveness and informational benefit. The concept was evaluated with respect to perception and distraction in an empirical dual-task study.

2 METHOD

2.1 Participants

Data from 24 participants (9 female, 15 male) were analyzed. Their ages ranged between 19 and 55 years, with a mean of 26.75 years ($SD = 7.08$ years). All participants had normal or corrected to normal vision. Three of them wore glasses during the experiment, while two used contact lenses. Four of the participants had prior experience with AR, such as with the Microsoft HoloLens or with smartphone-based AR.

2.2 Notification Concept

We developed three icon-based visualizations for notifications of increasing priority based on literature regarding peripheral vision and information presentation on smart glasses. To achieve the balance between perception and distraction, we aimed at increasing the stimulus intensity from low- to high-priority notifications. The low-priority notifications were of the lowest intensity to evoke the least distraction. The high-priority notifications were of high intensity which should be quicker to perceive while also leading to higher levels of distraction.

The exemplary icons we used are shown in Figure 1 and represent notifications from a public transport app. To increase the intensity, we used different animations, colors, and icon sizes. For the low-priority notification, we used the less peripheral-vision-friendly color green and avoided a sudden onset as the icon was first transparent and became gradually more opaque until it was completely opaque after five seconds. The medium-priority notification used the peripheral-vision-friendly colors yellow and blue and moved continuously up and down to achieve higher stimulus intensity. The high-priority notification used the signal color red and changed to yellow with a blue exclamation mark leading to a flashing and color-changing animation. Moreover, the size of the icon was slightly larger than the size of the icons from the previous categories to reach the highest intensity level.

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Figure 2: Example of the high-priority notification and the 2-back task as presented to the participants.

2.3 Study Design and Procedure

To evaluate our concept, we designed a within-subject study in which participants should react to the notifications while also performing a primary task in the central field of view of the smart glasses. We used three experiment blocks and in each block, one type of notification was displayed ten times. The order of the blocks was counterbalanced across participants. The notifications were presented in a display-fixed manner, on the left and right side randomly, and at an eccentricity of 20° (Figure 2). The interval between the notifications was randomized between 12–24 s. As primary task, we used the 2-back task. In this task, letters are presented one-by-one successively and the participant has to indicate for each one whether it matches the letter presented two items ago. Each block included 60 letters with 18 matches. As smart glasses, we used the *Nreal Light Developer Kit* with a field of view of 52°. Participants used the app button on the controller to react to notifications and the track pad to indicate matches in the 2-back task.

Each session took approximately 50 minutes and was conducted in a lab under constant lighting conditions. When the participants arrived, they first received oral and written instructions and filled in a demographic questionnaire. Afterwards, they performed a short training trial to familiarize themselves with their task and the *Nreal Light*. Then the three experiment blocks started. After each block, participants completed questionnaires regarding distraction and perception. Additionally, we collected their reaction times for the notifications and the 2-back-task. After the three experiment blocks, participants completed a concluding questionnaire including a ranking of the three notification types in terms of perceptibility and distraction.

3 RESULTS

With respect to the perception of the notifications, we had collected reaction time data as well as subjective ratings and rankings of the three notification types. The results showed that reaction times were considerably higher for the low-priority notification ($M = 2.44$ s, $SD = 0.65$ s) compared to notifications of medium ($M = 1.18$ s, $SD = 0.28$ s) and high priority ($M = 1.13$ s, $SD = 0.28$ s). A repeated-measures ANOVA with Greenhouse–Geisser correction revealed significant results, $F(2,46) = 88.23$, $p < 0.0001$, $\eta^2_G = 0.67$. Post-hoc tests with Bonferroni correction showed that only the pairwise comparisons between low- and medium-priority ($p < 0.0001$, $d = 2.16$) as well as low- and high-priority notifications ($p < 0.0001$, $d = 1.97$) were statistically significant with large effects. This was also reflected in the subjective ratings and rankings of the three notifications. The descriptive statistics showed that the high-priority notification received the best ratings and ranks regarding perceptibility followed closely by the medium-priority notification. The low-priority notification received lower ratings regarding perceptibility and was ranked third by 83.33% of participants.

To analyze the distraction evoked by the notifications, we analyzed the participants' performance in the 2-back task as well as ratings and rankings regarding distraction. The results for the performance in the 2-back task showed that the percentages of correct responses were at similar levels for all three notifications with means of 90.69% ($SD = 4.86\%$) for the low-priority, 87.85% ($SD = 6.03\%$) for the medium-priority, and 89.31% ($SD = 6.06\%$) for the high-priority notification. There were no statistically significant differences. The descriptive results from the subjective ratings indicated that the low-priority notifications were the least distracting, followed by the medium- and high-priority notifications. The ranking results showed a similar tendency.

4 DISCUSSION AND CONCLUSION

We can conclude that the results did not completely support our notification concept, mainly because there was no clear difference in distraction or perception between medium- and high-priority notifications. This may be due to certain limitations in our study design. For example, the flashing frequency of the high-priority notification could be improved and the design of the 2-back task may have additionally reduced the pop-out effect of the high-priority notification. The 2-back task may have been too easy as primary task. A more difficult task might result in clearer differences regarding the distracting effects of the notifications.

Nevertheless, the results indicate that it is possible to represent notifications on smart glasses with at least two priority levels. The low-priority notifications did show a trend towards lower distractibility while medium- and high-priority notifications were more distracting and perceived faster. Given that not all of these results reached statistical significance, further research is required to support our results and shed more light on open issues.

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